BLC's Roadmap For Coding (Don't Panic)

"Have Patience and Behave Like Compiler" - BLC

1. Logic Building

Tips for improving your programming logic:

Understand the problem: Read and understand the problem statement.
Generate Examples: Create additional input and output cases for each problem.
Draw observations: Draw observations and patterns based on the examples you created.
Start with Basic: First, think of the basic way to solve the problem then optimize the solution.

Basic Problems

Sl. No.	Problem	Reference Link	Status
1	Check Even or Odd	https://www.geeksforgeeks.org/dsa/check-whether-given-number-even-odd/	
2	Multiplication Table	https://www.geeksforgeeks.org/dsa/program-to-print-multiplication-table-of-a-number/	
3	Sum of Naturals	https://www.geeksforgeeks.org/dsa/program-find-sum-first-n-natural-numbers/	
4	Sum of Squares of Naturals	https://www.geeksforgeeks.org/dsa/sum-of-squares-of-first-n-natural-numbers/	
5	Swap Two Numbers	https://www.geeksforgeeks.org/dsa/swap-two-numbers/	
6	Closest Number	https://www.geeksforgeeks.org/dsa/find-number-closest-n-divisible-m/	
7	Dice Problem	https://www.geeksforgeeks.org/dsa/the-dice-problem/	
8	Nth Term of AP	https://www.geeksforgeeks.org/dsa/nth-term-of-ap-from-first-two-terms/	

Easy Problems

Sl. No.	Problem	Reference Link	Status
1	Sum of Digits	https://www.geeksforgeeks.org/dsa/program-for-sum-of-the-digits-of-a-given-number/	
2	Reverse Digits	https://www.geeksforgeeks.org/dsa/write-a-program-to-reverse-digits-of-a-number/	

3	Prime Testing	https://www.geeksforgeeks.org/dsa/introduction-to-primality-test-and-school-method/	
4	Check Power	https://www.geeksforgeeks.org/dsa/check-if-a-number-is-power-of-another-number/	
5	Distance between Two Points	https://www.geeksforgeeks.org/dsa/program-calculate-distance-two-points/	
6	Factorial of a Number	https://www.geeksforgeeks.org/dsa/program-for-factorial-of-a-number/	
7	GCD or HCF	https://www.geeksforgeeks.org/dsa/program-to-find-gcd-or-hcf-of-two-numbers/	
8	LCM of Two Numbers	https://www.geeksforgeeks.org/dsa/program-to-find-lcm-of-two-numbers/	
9	Perfect Number	https://www.geeksforgeeks.org/dsa/perfect-number/	
10	Day of the Week	https://www.geeksforgeeks.org/dsa/find-day-of-the-week-for-a-given-date/	
11	Nth Fibonacci Number	https://www.geeksforgeeks.org/dsa/program-for-nth-fibonacci-number/	
12	Decimal to Binary	https://www.geeksforgeeks.org/dsa/program-decimal-binary-conversion/	
13	N-th term of 1, 3, 6, 10, 15, 21…	https://www.geeksforgeeks.org/dsa/find-nth-term-series-136101521/	
14	Armstrong Number	https://www.geeksforgeeks.org/dsa/program-for-armstrong-numbers/	
15	Palindrome Number	https://www.geeksforgeeks.org/dsa/check-if-a-number-is-palindrome/	

Medium Problems

Sl. No.	Problem	Reference Link	Status
1	Square Root	https://www.geeksforgeeks.org/dsa/square-root-of-an-integer/	
2	3 Divisors	https://www.geeksforgeeks.org/dsa/numbers-exactly-3-divisors/	
3	Divisible by 4	https://www.geeksforgeeks.org/dsa/check-large-number-divisible-4-not/	
4	Divisibility by 11	https://www.geeksforgeeks.org/dsa/check-large-number-divisible-11-not/	
5	Divisibility by 13	https://www.geeksforgeeks.org/dsa/check-large-number-divisible-13-not/	
6	K-th Digit in a^b	https://www.geeksforgeeks.org/dsa/k-th-digit-raised-power-b/	
7	Compute nPr	https://www.geeksforgeeks.org/dsa/program-to-calculate-the-value-of-npr/	
8	Compute nCr	https://www.geeksforgeeks.org/dsa/program-calculate-value-ncr/	
9	Pascal’s Triangle	https://www.geeksforgeeks.org/dsa/pascal-triangle/	
10	All Factor (Or Divisors)	https://www.geeksforgeeks.org/dsa/find-all-factors-of-a-natural-number/	
11	Prime Factorization	https://www.geeksforgeeks.org/dsa/prime-factor/	
12	Largest Prime factor	https://www.geeksforgeeks.org/dsa/find-largest-prime-factor-number/	

Hard Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Sieve of Eratosthenes	https://www.geeksforgeeks.org/dsa/sieve-of-eratosthenes/	
2	Super Prime	https://www.geeksforgeeks.org/dsa/super-prime/	
3	Clock Angle Problem	https://www.geeksforgeeks.org/dsa/calculate-angle-hour-hand-minute-hand/	
4	Tower of Hanoi	https://www.geeksforgeeks.org/dsa/c-program-for-tower-of-hanoi/	
5	Rat and Poisoned	https://www.geeksforgeeks.org/dsa/rat-and-poisoned-bottle-problem/	
6	8 puzzle Problem	https://www.geeksforgeeks.org/dsa/8-puzzle-problem-using-branch-and-bound/	
7	Euler's Totient Function	https://www.geeksforgeeks.org/dsa/eulers-totient-function/	
8	Josephus Problem	https://www.geeksforgeeks.org/dsa/josephus-problem/	
9	Check if the given chessboard is valid or not	https://www.geeksforgeeks.org/dsa/check-if-the-given-chessboard-is-valid-or-not/	
10	Squares in a chessboard	https://www.geeksforgeeks.org/dsa/program-to-find-number-of-squares-on-a-chessboard/	

2. Time Complexity

Basic Study			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Why is Analysis Important?	https://www.geeksforgeeks.org/dsa/what-is-algorithm-and-why-analysis-of-it-is-important/	
2	Order of Growth	https://www.geeksforgeeks.org/gate/order-o-f-growth/	
3	Asymptotic Analysis	https://www.geeksforgeeks.org/dsa/asymptotic-notation-and-analysis-based-on-input-size-of-algorithms/	
4	Worst, Average and Best Cases	https://www.geeksforgeeks.org/dsa/worst-average-and-best-case-analysis-of-algorithms/	

Asymptotic Notations			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Big-O Notation	https://www.geeksforgeeks.org/dsa/analysis-algorithms-big-o-analysis/	
2	Theta - Θ Notation	https://www.geeksforgeeks.org/dsa/analysis-of-algorithms-big-theta-notation/	
3	Big – Ω (Big- Omega) Notation	https://www.geeksforgeeks.org/dsa/analysis-of-algorithms-big-omega-notation/	
4	Time Complexity	https://www.geeksforgeeks.org/dsa/understanding-time-complexity-simple-examples/	
5	Space Complexity	https://www.geeksforgeeks.org/dsa/g-fact-86/	
6	Big O vs, Big Omega vs Theta	https://www.geeksforgeeks.org/dsa/difference-between-big-oh-big-omega-and-big-theta/	
7	Examples of Big-O analysis	https://www.geeksforgeeks.org/dsa/examples-of-big-o-analysis/	
8	Practice Questions on Time Complexity Analysis	https://www.geeksforgeeks.org/dsa/practice-questions-time-complexity-analysis/	

Analysis Examples			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Analyzing Loops	https://www.geeksforgeeks.org/dsa/how-to-analyse-loops-for-complexity-analysis-of-algorithms/	
2	Analyzing Recursive Functions	https://www.geeksforgeeks.org/dsa/how-to-analyse-complexity-of-recurrence-relation/	
3	Amortized Analysis	https://www.geeksforgeeks.org/dsa/introduction-to-amortized-analysis/	

3. Arrays

Basics			
1	Introduction	https://www.geeksforgeeks.org/dsa/introduction-to-arrays-data-structure-and-algorithm-tutorials/	

Arrays in Different Languages			
Study according to your Language			
1	Arrays in C	https://www.geeksforgeeks.org/c/c-arrays/	
2	Vector in C++ STL	https://www.geeksforgeeks.org/cpp/vector-in-cpp-stl/	
3	Arrays in Java	https://www.geeksforgeeks.org/java/arrays-in-java/	
4	ArrayList in Java	https://www.geeksforgeeks.org/java/arraylist-in-java/	
5	List in Python	https://www.geeksforgeeks.org/python/python-lists/	

Basic Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Print Alternates	https://www.geeksforgeeks.org/dsa/print-alternate-elements-of-an-array/	
2	Leaders in an array	https://www.geeksforgeeks.org/dsa/leaders-in-an-array/	
3	Remove Duplicates from Sorted	https://www.geeksforgeeks.org/dsa/remove-duplicates-sorted-array/	
4	Generate all Subarrays	https://www.geeksforgeeks.org/dsa/generating-subarrays-using-recursion/	
5	Reverse an ArrayReverse an Array	https://www.geeksforgeeks.org/dsa/program-to-reverse-an-array/	
6	Rotate an Array	https://www.geeksforgeeks.org/dsa/complete-guide-on-array-rotations/	
7	Zeroes to End	https://www.geeksforgeeks.org/dsa/move-zeroes-end-array/	

Easy Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Duplicate within K Distance	https://www.geeksforgeeks.org/dsa/check-given-array-contains-duplicate-elements-within-k-distance/	
2	Sum of all Subarrays	https://www.geeksforgeeks.org/dsa/sum-of-all-subarrays/	
3	Stock Buy and Sell – Multiple Transactions	https://www.geeksforgeeks.org/dsa/stock-buy-sell/	
4	Missing Number	https://www.geeksforgeeks.org/dsa/find-the-missing-number/	
5	Max Subarray Sum	https://www.geeksforgeeks.org/dsa/largest-sum-contiguous-subarray/	
6	Two Sum - Find if there is a Pair	https://www.geeksforgeeks.org/dsa/check-if-pair-with-given-sum-exists-in-array/	
7	Two Sum - Closest Pair	https://www.geeksforgeeks.org/dsa/two-elements-whose-sum-is-closest-to-zero/	
8	Maximum Consecutive 1s with K Flips	https://www.geeksforgeeks.org/dsa/find-zeroes-to-be-flipped-so-that-number-of-consecutive-1s-is-maximized/	

9	Prefix Sum Array - Implementation	https://www.geeksforgeeks.org/dsa/prefix-sum-array-implementation-applications-competitive-programming/	

Pre-Requisities for Next Examples			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Binary Search	https://www.geeksforgeeks.org/dsa/binary-search/	
2	Selection Sort	https://www.geeksforgeeks.org/dsa/selection-sort-algorithm-2/	
3	Insertion Sort	https://www.geeksforgeeks.org/dsa/insertion-sort-algorithm/	
4	Quick Sort	https://www.geeksforgeeks.org/dsa/quick-sort-algorithm/	
5	Merge Sort	https://www.geeksforgeeks.org/dsa/merge-sort/	

4. Strings

Basics			
1	Introduction to Strings	https://www.geeksforgeeks.org/dsa/introduction-to-strings-data-structure-and-algorithm-tutorials/	

Strings in Different Languages			
Study according to your Language			
1	Strings in C	https://www.geeksforgeeks.org/c/strings-in-c/	
2	String Class in C++	https://www.geeksforgeeks.org/cpp/strings-in-cpp/	
3	String in Java	https://www.geeksforgeeks.org/java/strings-in-java/	
4	Python String	https://www.geeksforgeeks.org/python/python-string/	

Basic Operations			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Length of a String	https://www.geeksforgeeks.org/dsa/find-the-length-of-a-string/	
2	Check for Same	https://www.geeksforgeeks.org/dsa/program-to-check-if-two-strings-are-same-or-not/	

3	Search a Character	https://www.geeksforgeeks.org/dsa/program-to-search-a-character-in-a-string/	
4	Insert a Character	https://www.geeksforgeeks.org/dsa/how-to-insert-a-character-in-a-string/	
5	Remove a Character	https://www.geeksforgeeks.org/dsa/remove-a-character-from-a-given-position/	
6	Remove all occurrences	https://www.geeksforgeeks.org/dsa/remove-all-occurrences-of-a-character-in-a-string/	
7	Concatenating Two Strings	https://www.geeksforgeeks.org/computer-science-fundamentals/concatenation-of-two-strings/	
8	Reverse a string	https://www.geeksforgeeks.org/dsa/reverse-a-string/	
9	Generate all Substrings	https://www.geeksforgeeks.org/dsa/program-print-substrings-given-string/	

Easy Problems

Sl. No.	Problem	Reference Link	Status
1	Check for Binary	https://www.geeksforgeeks.org/dsa/check-for-binary-string/	
2	Camel Case Conversion	https://www.geeksforgeeks.org/dsa/camel-case-given-sentence/	
3	Substrings with Corners as 1s	https://www.geeksforgeeks.org/dsa/given-binary-string-count-number-substrings-start-end-1/	
4	Check for Panagram	https://www.geeksforgeeks.org/dsa/pangram-checking/	
5	Check for Palindrome	https://www.geeksforgeeks.org/dsa/palindrome-string/	
6	Check for Substring	https://www.geeksforgeeks.org/dsa/check-string-substring-another/	
7	Check for Subsequence	https://www.geeksforgeeks.org/dsa/given-two-strings-find-first-string-subsequence-second/	
8	Check for Anagrams	https://www.geeksforgeeks.org/dsa/check-whether-two-strings-are-anagram-of-each-other/	
9	Check for K-Anagrams	https://www.geeksforgeeks.org/dsa/check-two-strings-k-anagrams-not/	

Medium Problems

Sl. No.	Problem	Reference Link	Status
1	First Repeating Character	https://www.geeksforgeeks.org/dsa/find-the-first-repeated-character-in-a-string/	

2	First Non-Repeating Character	https://www.geeksforgeeks.org/dsa/given-a-string-find-its-first-non-repeating-character/	
3	Check for Rotation	https://www.geeksforgeeks.org/dsa/a-program-to-check-if-strings-are-rotations-of-each-other/	
4	K'th Non-repeating Character	https://www.geeksforgeeks.org/dsa/kth-non-repeating-character/	
5	Validate IP Address	https://www.geeksforgeeks.org/dsa/program-to-validate-an-ip-address/	
6	Isomorphic Strings	https://www.geeksforgeeks.org/dsa/check-if-two-given-strings-are-isomorphic-to-each-other/	
7	Roman to Integer	https://www.geeksforgeeks.org/dsa/roman-number-to-integer/	
8	Remove Adjacent	https://www.geeksforgeeks.org/dsa/recursively-remove-adjacent-duplicates-given-string/	
9	Longest Palindromic Substring	https://www.geeksforgeeks.org/dsa/longest-palindromic-substring/	
10	Anagram substrings	https://www.geeksforgeeks.org/dsa/count-total-anagram-substrings/	
11	Lexicographically next string	https://www.geeksforgeeks.org/dsa/lexicographically-next-string/	
12	Word Break Problem	https://www.geeksforgeeks.org/dsa/word-break-problem-dp-32/	

Two Pointer Technqiue

When to Use Two Pointers:

1. Sorted Input : If the array or list is already sorted (or can be sorted), two pointers can efficiently find pairs or ranges. Example: Find two numbers in a sorted array that

Two Pointer Technqiue (Easy Problems)

Sl. No.	Problem	Reference Link	Status
1	Remove Occurrences	https://www.geeksforgeeks.org/dsa/remove-element/	
2	Move Zeros To End	https://www.geeksforgeeks.org/dsa/move-zeroes-end-array/	
3	Unique Elements in Sorted Array	https://www.geeksforgeeks.org/dsa/remove-duplicates-sorted-array/	
4	Two Sum	https://www.geeksforgeeks.org/dsa/check-if-pair-with-given-sum-exists-in-array/	
5	Pair Sum in a Sorted and Rotated Array	https://www.geeksforgeeks.org/dsa/given-a-sorted-and-rotated-array-find-if-there-is-a-pair-with-a-given-sum/	
6	Closest Pair Sum	https://www.geeksforgeeks.org/dsa/2-sum-pair-sum-closest-to-target/	
7	Smallest Subarray > Sum	https://www.geeksforgeeks.org/dsa/minimum-length-subarray-sum-greater-given-value/	

8	Sentence Palindrome	https://www.geeksforgeeks.org/dsa/sentence-palindrome-palindrome-removing-spaces-dots-etc/	

Two Pointer Technqiue (Medium Problems)			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Count pairs with absolute difference equal to k	https://www.geeksforgeeks.org/dsa/count-pairs-difference-equal-k/	
2	Triplet Sum in Array	https://www.geeksforgeeks.org/dsa/find-a-triplet-that-sum-to-a-given-value/	
3	Sum of Two Equals Third	https://www.geeksforgeeks.org/dsa/find-triplet-sum-two-equals-third-element/	
4	Subarrays with Max in Range	https://www.geeksforgeeks.org/dsa/number-subarrays-maximum-value-given-range/	
5	Longest Substring with K Unique	https://www.geeksforgeeks.org/dsa/find-the-longest-substring-with-k-unique-characters-in-a-given-string/	
6	The Celebrity Problem	https://www.geeksforgeeks.org/dsa/the-celebrity-problem/	

Two Pointer Technqiue (Hard Problems)			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Trapping Rain Water Problem	https://www.geeksforgeeks.org/dsa/trapping-rain-water/	
2	Kadane's Algorithm	https://www.geeksforgeeks.org/dsa/largest-sum-contiguous-subarray/	

Prefix Sum (Easy Problems)			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Equilibrium Index	https://www.geeksforgeeks.org/dsa/equilibrium-index-of-an-array/	
2	Split an array into two equal Sum subarrays	https://www.geeksforgeeks.org/dsa/split-array-two-equal-sum-subarrays/	
3	Mean of range in array	https://www.geeksforgeeks.org/dsa/mean-range-array/	
4	Original Array from Prefix Sums	https://www.geeksforgeeks.org/dsa/find-original-array-from-given-prefix-sum-array/	

Prefix Sum (Medium and Hard Problems)			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Product of Array Except Self	https://www.geeksforgeeks.org/dsa/a-product-array-puzzle/	
2	Longest Subarray With Sum K	https://www.geeksforgeeks.org/dsa/longest-sub-array-sum-k/	

3	Subarray with 0 sum	https://www.geeksforgeeks.org/dsa/find-if-there-is-a-subarray-with-0-sum/	
4	Subarray Sum Divisible By K	https://www.geeksforgeeks.org/dsa/count-sub-arrays-sum-divisible-k/	
5	Count Subarrays having Sum K	https://www.geeksforgeeks.org/dsa/number-subarrays-sum-exactly-equal-k/	
6	Smallest sum contiguous subarray	https://www.geeksforgeeks.org/dsa/smallest-sum-contiguous-subarray/	

Hashing Data Structure			
1	Introduction	https://www.geeksforgeeks.org/dsa/introduction-to-hashing-2/	
2	Applications	https://www.geeksforgeeks.org/dsa/applications-of-hashing/	

Easy Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Check tor Subset	https://www.geeksforgeeks.org/dsa/find-whether-an-array-is-subset-of-another-array-set-1/	
2	Check for Disjoint	https://www.geeksforgeeks.org/dsa/check-two-given-sets-disjoint/	
3	Fizz Buzz	https://www.geeksforgeeks.org/dsa/fizz-buzz-implementation/	
4	Max distance between two occurrences	https://www.geeksforgeeks.org/dsa/maximum-distance-two-occurrences-element-array/	
5	Intersection of Two Arrays	https://www.geeksforgeeks.org/dsa/intersection-of-two-arrays/	
6	Union of Two Arrays	https://www.geeksforgeeks.org/dsa/union-of-two-arrays/	
7	2 Sum - Count Pairs	https://www.geeksforgeeks.org/dsa/count-pairs-with-given-sum/	

Medium Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Pair Sums Divisible by k	https://www.geeksforgeeks.org/dsa/check-if-an-array-can-be-divided-into-pairs-whose-sum-is-divisible-by-k/	
2	3 Sum - Count all triplets with Given sum	https://www.geeksforgeeks.org/dsa/3-sum-count-all-triplets-with-given-sum/	
3	Longest Subarray with Majority Greater Than K	https://www.geeksforgeeks.org/dsa/length-of-longest-subarray-in-which-elements-greater-than-k-are-more-than-elements-not-greater-than-k/	
4	Subarray with given sum	https://www.geeksforgeeks.org/dsa/find-subarray-with-given-sum-in-array-of-integers/	

5	Longest Consecutive subsequence	https://www.geeksforgeeks.org/dsa/count-minimum-number-subsets-subsequences-consecutive-numbers/	
6	Distincts in every window of size k	https://www.geeksforgeeks.org/dsa/count-distinct-elements-in-every-window-of-size-k/	
7	Smallest subarray with k distinct numbers	https://www.geeksforgeeks.org/dsa/smallest-subarray-k-distinct-numbers/	
8	All pairs (a, b) in an array such that a % b = k	https://www.geeksforgeeks.org/dsa/find-pairs-b-array-b-k/	

Hard Problems

Sl. No.	Problem	Reference Link	Status
1	Sum of unique sub-array sums	https://www.geeksforgeeks.org/dsa/find-sum-unique-sub-array-sum-given-array/	
2	Array Leaders	https://www.geeksforgeeks.org/problems/leaders-in-an-array-1587115620/1?page=1&sortBy=submissions	
3	Majority Element	https://www.geeksforgeeks.org/problems/majority-element-1587115620/1?page=1&sortBy=submissions	
4	kth Smallest	https://www.geeksforgeeks.org/problems/kth-smallest-element5635/1?page=1&sortBy=submissions	
5	Anagram	https://www.geeksforgeeks.org/problems/anagram-1587115620/1?page=3&sortBy=submissions	
6	Palindrome String	https://www.geeksforgeeks.org/problems/palindrome-string0817/1?page=3&sortBy=submissions	
7	Largest Array with SUm 0	https://www.geeksforgeeks.org/problems/largest-subarray-with-0-sum/1?page=3&sortBy=submissions	
8	Coin Change (Count Ways)	https://www.geeksforgeeks.org/problems/coin-change2448/1?page=6&sortBy=submissions	
9	Find the number of islands	https://www.geeksforgeeks.org/problems/find-the-number-of-islands/1?page=8&sortBy=submissions	

Sliding Window

Sl. No.	Content	Reference Link	Status
1	Introduction	https://www.geeksforgeeks.org/dsa/window-sliding-technique/	

Easy Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Maximum sum of a subarray of size k	https://www.geeksforgeeks.org/dsa/find-maximum-minimum-sum-subarray-size-k/	
2	Smallest window containing 0, 1 and 2	https://www.geeksforgeeks.org/dsa/smallest-window-containing-0-1-and-2/	
3	Count Strictly Increasing Subarrays	https://www.geeksforgeeks.org/dsa/count-strictly-increasing-subarrays/	

Medium Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Longest substring with distinct characters	https://www.geeksforgeeks.org/dsa/length-of-the-longest-substring-without-repeating-characters/	
2	Substrings with K Distinct	https://www.geeksforgeeks.org/dsa/count-number-of-substrings-with-exactly-k-distinct-characters/	
3	Maximum Fruits in Two Baskets	https://www.geeksforgeeks.org/dsa/maximum-number-of-fruits-in-two-baskets/	
4	Minimum Removals for Target Sum	https://www.geeksforgeeks.org/dsa/minimum-removals-for-target-sum/	
5	Binary subarray with sum	https://www.geeksforgeeks.org/dsa/count-of-subarrays-with-sum-equals-k-in-given-binary-array/	
6	Subarrays Product Less than K	https://www.geeksforgeeks.org/dsa/number-subarrays-product-less-k/	
7	Smallest window that contains all characters of string itself	https://www.geeksforgeeks.org/dsa/smallest-window-contains-characters-string/	
8	Smallest window in a String containing all characters of other String	https://www.geeksforgeeks.org/dsa/find-the-smallest-window-in-a-string-containing-all-characters-of-another-string/	

Medium Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Maximum of minimum for every window size	https://www.geeksforgeeks.org/dsa/find-the-maximum-of-minimums-for-every-window-size-in-a-given-array/	
2	Longest Substring with K Uniques	https://www.geeksforgeeks.org/dsa/find-the-longest-substring-with-k-unique-characters-in-a-given-string/	

3	Minimum Window Substring	https://www.geeksforgeeks.org/problems/smallest-window-in-a-string-containing-all-the-characters-of-another-string-1587115621/1	

5. Recursion

Basics			
1	Introduction	https://www.geeksforgeeks.org/dsa/introduction-to-recursion-2/	
2	Tail Recursion	https://www.geeksforgeeks.org/dsa/tail-recursion/	
3	Tail Recursion Elimination	https://www.geeksforgeeks.org/dsa/tail-call-elimination/	

Recursion In Different Languages			
1	Recursion in Python	https://www.geeksforgeeks.org/python/recursion-in-python/	
2	Recursion in Java	https://www.geeksforgeeks.org/java/recursion-in-java/	
3	C Recursion	https://www.geeksforgeeks.org/c/c-recursion/	
4	C++ Recursion	https://www.geeksforgeeks.org/cpp/cpp-recursion/	

Easy Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Factorial	https://www.geeksforgeeks.org/dsa/program-for-factorial-of-a-number/	
2	Print 1 to n	https://www.geeksforgeeks.org/dsa/print-n-to-1-without-loop/	
3	Print n to 1	https://www.geeksforgeeks.org/dsa/print-n-to-1-without-loop/	
4	Sum of array	https://www.geeksforgeeks.org/dsa/sum-array-elements-using-recursion/	
5	Reverse a string	https://www.geeksforgeeks.org/dsa/reverse-a-string-using-recursion/	
6	Decimal to binary	https://www.geeksforgeeks.org/dsa/decimal-binary-number-using-recursion/	
7	Sum of digit	https://www.geeksforgeeks.org/dsa/sum-digit-number-using-recursion/	
8	Array Min and Max	https://www.geeksforgeeks.org/dsa/recursive-programs-to-find-minimum-and-maximum-elements-of-array/	
9	Palindrome Check	https://www.geeksforgeeks.org/dsa/recursive-function-check-string-palindrome/	

Medium Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Mean of Array	https://www.geeksforgeeks.org/dsa/mean-of-array-using-recursion/	
2	Adjacent duplicates	https://www.geeksforgeeks.org/dsa/mean-of-array-using-recursion/	
3	Coin Change	https://www.geeksforgeeks.org/dsa/coin-change-dp-7/	
4	Longest palindromic substring	https://www.geeksforgeeks.org/dsa/length-of-longest-palindromic-sub-string-recursion/	
5	Tower of Hanoi	https://www.geeksforgeeks.org/dsa/c-program-for-tower-of-hanoi/	
6	Permutations	https://www.geeksforgeeks.org/dsa/write-a-c-program-to-print-all-permutations-of-a-given-string/	
7	Subsets	https://www.geeksforgeeks.org/dsa/backtracking-to-find-all-subsets/	

Hard Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Sort a Queue	https://www.geeksforgeeks.org/dsa/sort-the-queue-using-recursion/	
2	Palindromic Partitions	https://www.geeksforgeeks.org/dsa/given-a-string-print-all-possible-palindromic-partition/	
3	Scrambled Strings	https://www.geeksforgeeks.org/dsa/check-if-a-string-is-a-scrambled-form-of-another-string/	
4	N Queen Problem	https://www.geeksforgeeks.org/dsa/n-queen-problem-backtracking-3/	
5	Sudoku Solver	https://www.geeksforgeeks.org/dsa/sudoku-backtracking-7/	
6	Knight's Tour	https://www.geeksforgeeks.org/dsa/the-knights-tour-problem/	

6. Greedy Algorithm

Greedy algorithms are a class of algorithms that make locally optimal choices at each step with the hope of finding a global optimum solution.

At every step of the algorithm, we make a choice that looks the best at the moment. To make the choice, we sometimes sort the array so that we can always get the next optimal choice quickly. We sometimes also use a priority queue to get the next optimal item.
After making a choice, we check for constraints (if there are any) and keep picking until we find the solution.
Greedy algorithms do not always give the best solution. For example, in coin change and 0/1 knapsack problems, we get the best solution using Dynamic Programming.
Examples of popular algorithms where Greedy gives the best solution are Fractional Knapsack, Dijkstra's algorithm, Kruskal's algorithm, Huffman coding and Prim's Algorithm

Basics			
1	Introduction	https://www.geeksforgeeks.org/dsa/introduction-to-greedy-algorithm-data-structures-and-algorithm-tutorials/	
2	General Structure	https://www.geeksforgeeks.org/dsa/greedy-algorithms-general-structure-and-applications/	

Easy Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Fractional Knapsack	https://www.geeksforgeeks.org/dsa/fractional-knapsack-problem/	
2	Min Cost to Make Array Size 1	https://www.geeksforgeeks.org/dsa/minimum-cost-make-array-size-1-removing-larger-pairs/	
3	Smallest Subset Greater Sum	https://www.geeksforgeeks.org/dsa/smallest-subset-sum-greater-elements/	
4	Assign Cookies	https://www.geeksforgeeks.org/dsa/assign-cookies/	
5	Buy Maximum Stocks	https://www.geeksforgeeks.org/dsa/buy-maximum-stocks-stocks-can-bought-th-day/	
6	Max Consecutive Diff Sum	https://www.geeksforgeeks.org/dsa/maximize-sum-consecutive-differences-circular-array/	

7	Min and Max Costs to buy all	https://www.geeksforgeeks.org/dsa/find-minimum-maximum-amount-buy-n-candies/	

Medium Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Activity Selection	https://www.geeksforgeeks.org/dsa/activity-selection-problem-greedy-algo-1/	
2	Jump Game	https://www.geeksforgeeks.org/dsa/minimum-number-jumps-reach-endset-2on-solution/	
3	Max trains	https://www.geeksforgeeks.org/dsa/maximum-trains-stoppage-can-provided/	
4	Paper cut into min squares	https://www.geeksforgeeks.org/dsa/paper-cut-minimum-number-squares/	
5	Max Satisfied Customers	https://www.geeksforgeeks.org/dsa/maximum-number-customers-can-satisfied-given-quantity/	
6	Water Connection Problem	https://www.geeksforgeeks.org/dsa/water-connection-problem/	
7	Policemen catch thieves	https://www.geeksforgeeks.org/dsa/policemen-catch-thieves/	
8	Assign Mice to Holes	https://www.geeksforgeeks.org/dsa/assign-mice-holes/	

7. Dynamic Programming or DP

Dynamic Programming is an algorithmic technique with the following properties.

It is mainly an optimization over plain recursion. Wherever we see a recursive solution that has repeated calls for the same inputs, we can optimize it using Dynamic Programming.
The idea is to simply store the results of subproblems so that we do not have to re-compute them when needed later. This simple optimization typically reduces time complexities from exponential to polynomial.
Some popular problems solved using Dynamic Programming are Fibonacci Numbers, Diff Utility (Longest Common Subsequence), Bellman–Ford Shortest Path, Floyd Warshall, Edit Distance and Matrix Chain Multiplication.

Basics			
1	Introduction	https://www.geeksforgeeks.org/dsa/introduction-to-dynamic-programming-data-structures-and-algorithm-tutorials/	

2	Tabulation vs Memoization	https://www.geeksforgeeks.org/dsa/tabulation-vs-memoization/	
3	Steps to solve a DP Problem	Steps to solve a DP Problem	

Basic Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Fibonacci numbers	https://www.geeksforgeeks.org/dsa/program-for-nth-fibonacci-number/	
2	Tribonacci Numbers	https://www.geeksforgeeks.org/dsa/tribonacci-numbers/	
3	Climbing Stairs	https://www.geeksforgeeks.org/dsa/count-ways-reach-nth-stair/	
4	Climbing Stairs with 3 Moves	https://www.geeksforgeeks.org/dsa/count-ways-reach-nth-stair-using-step-1-2-3/	
5	nth Catalan Number	https://www.geeksforgeeks.org/dsa/program-nth-catalan-number/	
6	Count Valid Parenthesis	https://www.geeksforgeeks.org/dsa/find-number-valid-parentheses-expressions-given-length/	

Easy Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	House Robber	https://www.geeksforgeeks.org/dsa/find-maximum-possible-stolen-value-houses/	
2	Min Cost Path	https://www.geeksforgeeks.org/dsa/min-cost-path-dp-6/	
3	Subset Sum Problem	https://www.geeksforgeeks.org/dsa/subset-sum-problem-dp-25/	
4	Coin change problem - Count Ways	https://www.geeksforgeeks.org/dsa/coin-change-dp-7/	
5	Coin Change – Minimum Coins to Make Sum	https://www.geeksforgeeks.org/dsa/find-minimum-number-of-coins-that-make-a-change/	
6	Cutting a Rod	https://www.geeksforgeeks.org/dsa/cutting-a-rod-dp-13/	
7	Longest Common Substring	https://www.geeksforgeeks.org/dsa/longest-common-substring-space-optimized-dp-solution/	

Medium Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Water Overflow	https://www.geeksforgeeks.org/dsa/find-water-in-a-glass/	

2	Longest Common Subsequence	https://www.geeksforgeeks.org/dsa/longest-common-subsequence-dp-4/	
3	Longest Increasing Subsequence	https://www.geeksforgeeks.org/dsa/longest-increasing-subsequence-dp-3/	
4	Tile Stacking Problem	https://www.geeksforgeeks.org/dsa/tile-stacking-problem/	
5	Longest Palindromic Subsequence	https://www.geeksforgeeks.org/dsa/longest-palindromic-subsequence-dp-12/	
6	Longest Common Increasing Subsequence (LCS + LIS)	https://www.geeksforgeeks.org/dsa/longest-common-increasing-subsequence-lcs-lis/	
7	Count Derangements	https://www.geeksforgeeks.org/dsa/count-derangements-permutation-such-that-no-element-appears-in-its-original-position/	

Hard Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Egg Dropping Problem	https://www.geeksforgeeks.org/dsa/egg-dropping-puzzle-dp-11/	
2	Palindromic Substring Count	https://www.geeksforgeeks.org/dsa/count-palindrome-sub-strings-string/	
3	Stock Buy and Sell - At-Most k Times	https://www.geeksforgeeks.org/dsa/maximum-profit-by-buying-and-selling-a-share-at-most-twice/	
4	Stock Buy and Sell - At Most 2 Times	https://www.geeksforgeeks.org/dsa/maximum-profit-by-buying-and-selling-a-share-at-most-twice/	
5	Longest repeating and non-overlapping substring	https://www.geeksforgeeks.org/dsa/longest-repeating-and-non-overlapping-substring/	

8. Bitwise Algorithms

Basics			
1	Introduction to Bitwise Algorithms	https://www.geeksforgeeks.org/dsa/introduction-to-bitwise-algorithms-data-structures-and-algorithms-tutorial/	

In Different languages			
1	Bitwise Operators in C/C++	https://www.geeksforgeeks.org/c/bitwise-operators-in-c-cpp/	
2	Bitwise Operators in Java	https://www.geeksforgeeks.org/java/bitwise-operators-in-java/	
3	Python Bitwise Operators	https://www.geeksforgeeks.org/python/python-bitwise-operators/	

Bit Manipulation Tips and Tricks			
1	Bits manipulation (Important tactics)	https://www.geeksforgeeks.org/dsa/bits-manipulation-important-tactics/	

Medium Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Binary Representation	https://www.geeksforgeeks.org/dsa/binary-representation-of-a-given-number/	
2	Turn off the rightmost set bit	https://www.geeksforgeeks.org/dsa/turn-off-the-rightmost-set-bit/	
3	Check if K-th Bit Set	https://www.geeksforgeeks.org/dsa/check-whether-k-th-bit-set-not/	
4	Set the K-th Bit	https://www.geeksforgeeks.org/dsa/set-k-th-bit-given-number/	
5	Odd Occurring Number	https://www.geeksforgeeks.org/dsa/find-the-number-occurring-odd-number-of-times/	
6	Power of two	https://www.geeksforgeeks.org/dsa/program-to-find-whether-a-given-number-is-power-of-2/	
7	Check for opposite signs	https://www.geeksforgeeks.org/dsa/detect-if-two-integers-have-opposite-signs/	
8	Russian Peasant	https://www.geeksforgeeks.org/dsa/russian-peasant-multiply-two-numbers-using-bitwise-operators/	
9	Swap Two Numbers	https://www.geeksforgeeks.org/dsa/swap-two-numbers-without-using-temporary-variable/	

Medium Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Most Significant Set Bit	https://www.geeksforgeeks.org/dsa/find-significant-set-bit-number/	
2	Rightmost Set Bit	https://www.geeksforgeeks.org/dsa/position-of-rightmost-set-bit/	
3	Count Set Bits	https://www.geeksforgeeks.org/dsa/position-of-rightmost-set-bit/	
4	Swap Bits	https://www.geeksforgeeks.org/dsa/count-set-bits-in-an-integer/	
5	Rotate Bits	https://www.geeksforgeeks.org/dsa/rotate-bits-of-an-integer/	

6	Smallest power of 2 greater than or equal to n	https://www.geeksforgeeks.org/dsa/smallest-power-of-2-greater-than-or-equal-to-n/	
7	Check if binary is palindrome	https://www.geeksforgeeks.org/dsa/check-binary-representation-number-palindrome/	

9. Back Tracking Algorithms

Basics			
1	Introduction	https://www.geeksforgeeks.org/dsa/introduction-to-backtracking-2/	
2	Backtracking vs Branch and Bound	https://www.geeksforgeeks.org/dsa/difference-between-backtracking-and-branch-n-bound-technique/	

Standard Problems			
Sl. No.	**Problem**	**Reference Link**	**Status**
1	Permutations of a String	https://www.geeksforgeeks.org/dsa/write-a-c-program-to-print-all-permutations-of-a-given-string/	
2	The Knight's tour problem	https://www.geeksforgeeks.org/dsa/the-knights-tour-problem/	
3	Rat in a Maze	https://www.geeksforgeeks.org/dsa/rat-in-a-maze/	
4	N Queen Problem	https://www.geeksforgeeks.org/dsa/n-queen-problem-backtracking-3/	
5	Find all subsets	https://www.geeksforgeeks.org/dsa/backtracking-to-find-all-subsets/	
6	Tug of War	https://www.geeksforgeeks.org/dsa/tug-of-war/	